

Lecture 7: Pandas continued

CMSC/DATA 320: Introduction to Data Science

Lecturer: Anna Evtushenko (annae@umd.edu)

Announcements

- WA1, PA1, Quiz 3 grades released today
- PA1, Quiz 3 solutions released today
- PA2, WA2 due this Sunday

4. GROUP BY

Group tuples together by column/dimension

ID	A	B	C
1	foo	3	6.6
2	bar	2	4.7
3	foo	4	3.1
4	foo	3	8.0
5	bar	1	1.2
6	bar	2	2.5
7	foo	4	2.3
8	foo	3	8.0



A = "foo"

ID	B	C
1	3	6.6
3	4	3.1
4	3	8.0
7	4	2.3
8	3	8.0

A = "bar"

ID	B	C
2	2	4.7
5	1	1.2
6	2	2.5

4. GROUP BY

Group tuples together by column/dimension

ID	A	B	C
1	foo	3	6.6
2	bar	2	4.7
3	foo	4	3.1
4	foo	3	8.0
5	bar	1	1.2
6	bar	2	2.5
7	foo	4	2.3
8	foo	3	8.0

By 'B' →

B = 1

ID	A	C
5	bar	1.2

B = 2

ID	A	C
2	bar	4.7
6	bar	2.5

B = 3

ID	A	C
1	foo	6.6
4	foo	8.0
8	foo	8.0

B = 4

ID	A	C
3	foo	3.1
7	foo	2.3

4. GROUP BY

Group tuples together by column/dimension

ID	A	B	C
1	foo	3	6.6
2	bar	2	4.7
3	foo	4	3.1
4	foo	3	8.0
5	bar	1	1.2
6	bar	2	2.5
7	foo	4	2.3
8	foo	3	8.0

By 'A', 'B'



A = "bar", B = 1

ID	C
5	1.2

A = "bar", B = 2

ID	C
2	4.7
6	2.5

A = "foo", B = 3

ID	C
1	6.6
4	8.0
8	8.0

A = "foo", B = 4

ID	C
3	3.1
7	2.3

5. GROUP BY AGGREGATE

Compute one aggregate per group

ID	A	B	C
1	foo	3	6.6
2	bar	2	4.7
3	foo	4	3.1
4	foo	3	8.0
5	bar	1	1.2
6	bar	2	2.5
7	foo	4	2.3
8	foo	3	8.0

Group by 'B'
Sum on C

B = 1

ID	A	C
5	bar	1.2

B = 2

ID	A	C
2	bar	4.7
6	bar	2.5

B = 3

ID	A	C
1	foo	6.6
4	foo	8.0
8	foo	8.0

B = 4

ID	A	C
3	foo	3.1
7	foo	2.3

B = 1

Sum (C)
1.2

B = 2

Sum (C)
7.2

B = 3

Sum (C)
22.6

B = 4

Sum (C)
5.4

5. GROUP BY AGGREGATE

Final result usually seen as a table

ID	A	B	C
1	foo	3	6.6
2	bar	2	4.7
3	foo	4	3.1
4	foo	3	8.0
5	bar	1	1.2
6	bar	2	2.5
7	foo	4	2.3
8	foo	3	8.0

Group by 'B'
Sum on C

B = 1

Sum (C)
1.2

B = 2

Sum (C)
7.2

B = 3

Sum (C)
22.6

B = 4

Sum (C)
5.4



B	SUM(C)
1	1.2
2	7.2
3	22.6
4	5.4

6. UNION / INTERSECTION / DIFFERENCE

Set operations – only if the two tables have identical attributes/columns

ID	A	B	C
1	foo	3	6.6
2	bar	2	4.7
3	foo	4	3.1
4	foo	3	8.0

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ID	A	B	C
5	bar	1	1.2
6	bar	2	2.5
7	foo	4	2.3
8	foo	3	8.0



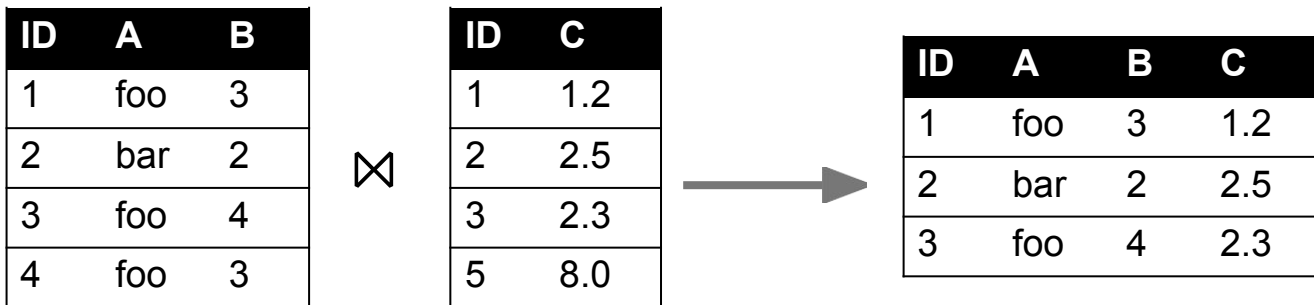
ID	A	B	C
1	foo	3	6.6
2	bar	2	4.7
3	foo	4	3.1
4	foo	3	8.0
5	bar	1	1.2
6	bar	2	2.5
7	foo	4	2.3
8	foo	3	8.0

Similarly Intersection (returns the rows that are common to both tables) and Set Difference EXCEPT/MINUS (returns the rows that are in one table but not in the other) manipulate tables as Sets

The way IDs are treated (whether as unique or non-unique identifiers) can lead to different behaviors in these operations

7. MERGE OR JOIN

Combine rows/tuples across two tables if they have the same key



What about IDs not present in both tables?

Often need to keep them around

Can “pad” with NaN

JOINING

DATA

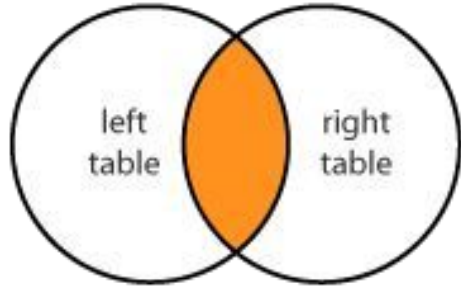
A join operation merges two or more tables into a single relation. Different ways of doing this:

- Inner - Returns only the rows where there is a match in **both** tables.
- Left - Returns all rows from the **left table**, and matching rows from the right table.
- Right- Returns all rows from the **right table**, and matching rows from the left table.
- Full Outer- Returns all rows when there is a match in **either** table.

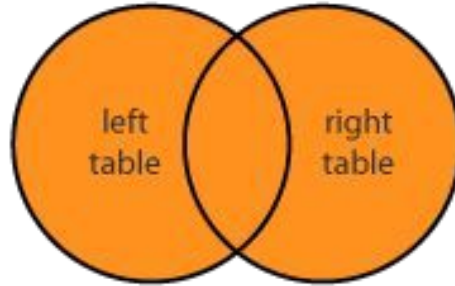
Join operations are done on columns that explicitly link the tables together

TYPES OF JOINS

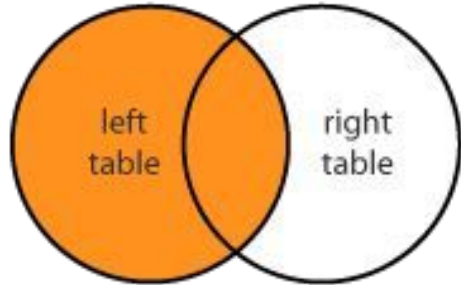
INNER JOIN



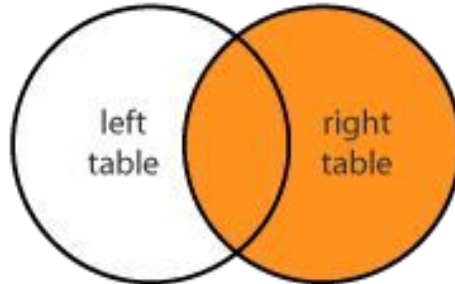
FULL JOIN



LEFT JOIN



RIGHT JOIN



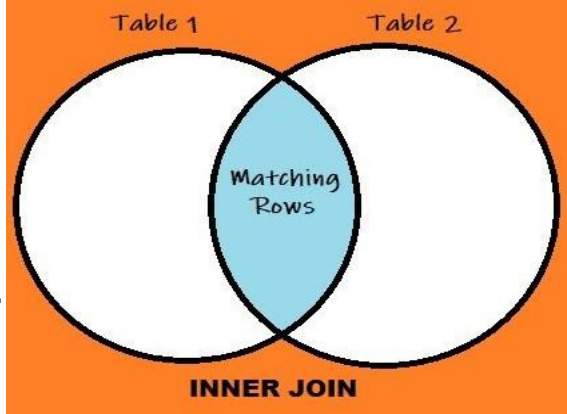
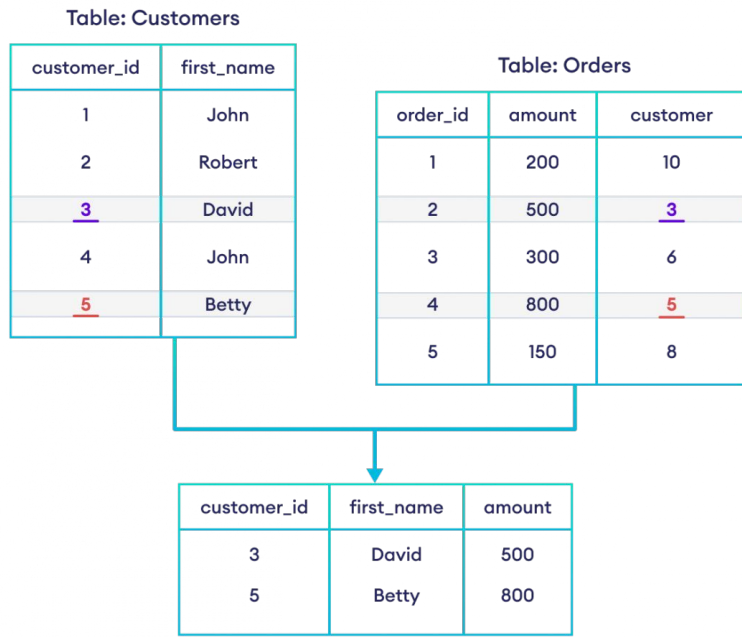
- **INNER JOIN:** Returns only rows with matches in both tables.
- **FULL JOIN:** Returns all rows from both tables, with NULLs where there are no matches.
- **LEFT JOIN:** Returns all rows from the left table, with NULLs for no matches on the right.
- **RIGHT JOIN:** Returns all rows from the right table, with NULLs for no matches on the left.

1. Inner Join

Returns only the rows where there is a match in both tables.

- Non-matching rows are excluded.

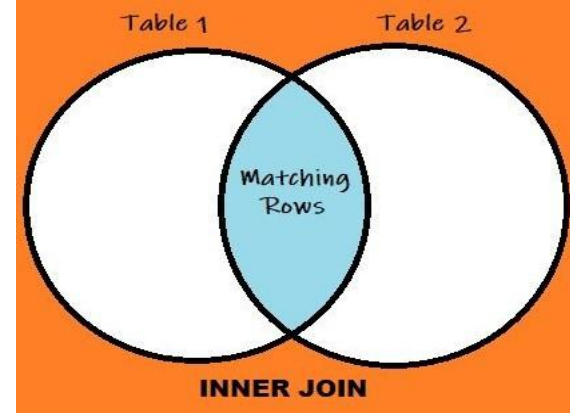
Example use: Get customers who placed an order.



How a query is structured (INNER JOIN)

Returns only the rows with matching values in both tables.

- INNER join **compares** each row of the first table with each row of the second table, to **find all pairs of rows** that satisfy the join-predicate.
- When the join-predicate is satisfied, the column values from both tables are combined into a new table.



EmpDetails			MaritalStatus		
ID	Name	Salary	ID	Name	Status
1	John	40000	1	John	Married
2	Alex	25000	3	Simon	Married
3	Simon	43000	4	Stella	Unmarried

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ID	Name	Salary	Status
1	John	40000	Married
3	Simon	43000	Married

INNER JOIN selects only **matching left and right table rows**.

More example: INNER JOINS

id	name
1	Megabyte
2	Meowly Cyrus
3	Fuzz Aldrin
4	Chairman Meow
5	Anderson Pooper
6	Gigabyte

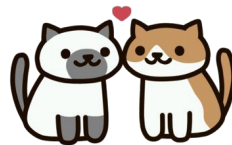
cats

cat_id	last_visit
1	02-16-2017
2	02-14-2017
5	02-03-2017

visits

Inner join returns merged rows that share the same value in the column they are being joined on (`id` and `cat_id`).

id	name	last_visit
1	Megabyte	02-16-2017
2	Meowly Cyrus	02-14-2017
5	Anderson Pooper	02-03-2017



LEFT JOINS

Inner joins are the most common type of joins (get results that appear in **both** tables)

Left joins: all the results from the left table, only **some matching results from the right table**

Given 2
Tables:

id	name
1	Megabyte
2	Meowly Cyrus
3	Fuzz Aldrin
4	Chairman Meow
5	Anderson Pooper
6	Gigabyte

cat_id	last_visit
1	02-16-2017
2	02-14-2017
5	02-03-2017

visits

Returns all rows from the **left table**, and matching rows from the right table.

If there's no match, you'll see **NULL** values from the right table.

Example use: Get all customers, whether or not they placed an order.

LEFT JOINS

Given 2 Tables:

id	name
1	Megabyte
2	Meowly Cyrus
3	Fuzz Aldrin
4	Chairman Meow
5	Anderson Pooper
6	Gigabyte

cat_id	last_visit
1	02-16-2017
2	02-14-2017
5	02-03-2017

visits

id	name	last_visit
1	Megabyte	02-16-2017
2	Meowly Cyrus	02-14-2017
3	Fuzz Aldrin	NULL
4	Chairman Meow	NULL
5	Anderson Pooper	02-03-2017
6	Gigabyte	NULL

Returns all rows from the **left table**, and matching rows from the right table.

If there's no match, you'll see **NULL** values from the right table.

Example use: Get all customers, whether or not they placed an order.

RIGHT JOINS

id	name
1	Megabyte
2	Meowly Cyrus
3	Fuzz Aldrin
4	Chairman Meow
5	Anderson Pooper
6	Gigabyte

cats

cat_id	last_visit
1	02-16-2017
2	02-14-2017
5	02-03-2017
7	02-19-2017
12	02-21-2017

visits

Right Join (or Right Outer Join)

- Returns all rows from the **right table**, and matching rows from the left table.
- If there's no match, you'll see **NULL** values from the left table.
- Example use: Get all orders, whether or not they are linked to a customer record.

id	name	last_visit
1	Megabyte	02-16-2017
2	Meowly Cyrus	02-14-2017
5	Anderson Pooper	02-03-2017
7	NULL	02-19-2017
12	NULL	02-21-2017

FULL OUTER JOIN

Returns all rows from both tables, with NULLs where there are no matches.

Given 2
Tables:

id	name
1	Megabyte
2	Meowly Cyrus
3	Fuzz Aldrin
4	Chairman Meow
5	Anderson Pooper
6	Gigabyte

cat_id	last_visit
1	02-16-2017
2	02-14-2017
5	02-03-2017

What do we get for an outer join?

Pandas demo